

DBMS – PURPLE COMPANY SELLER AND INVENTORY MANAGEMENT

Purple Company – Beauty Products E-Commerce Order Management System

1. Database Design:

1.1 Seller Table:

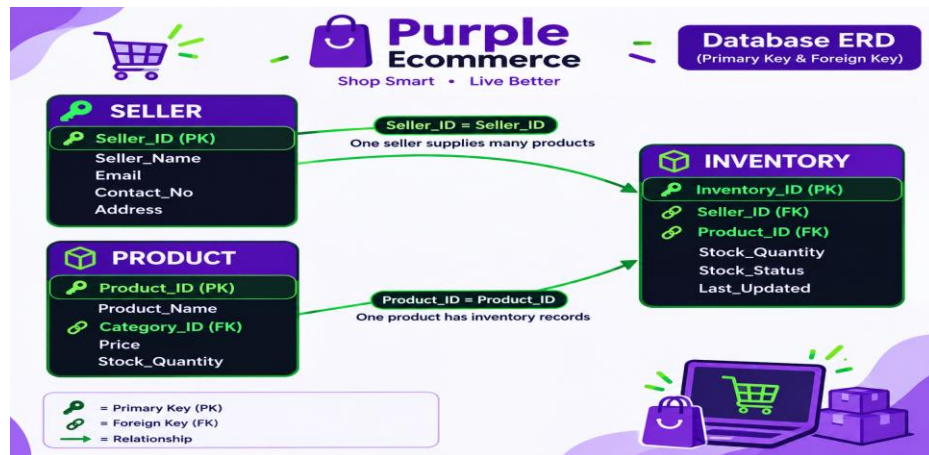
Attribute	Data Type	Constraint
Seller_ID	NUMBER	Primary Key
Seller_Name	VARCHAR2(100)	NOT NULL
Email	VARCHAR2(100)	NOT NULL, UNIQUE
Contact_No	VARCHAR2(15)	NOT NULL, UNIQUE
Address	VARCHAR2(200)	NOT NULL

1.2 Inventory Table:

Attribute	Data Type	Constraint
Inventory_ID	NUMBER	Primary Key
Product_ID	NUMBER	Foreign Key
Seller_ID	NUMBER	Foreign Key
Stock_Quantity	NUMBER	NOT NULL
Stock_Status	VARCHAR2(20)	NOT NULL
Last_Updated	DATE	NOT NULL

2. Table Relationships:

- PurpleSeller stores seller details.
- PurpleInventory stores stock details.
- Seller_ID in PurpleInventory references PurpleSeller.
- Product_ID in PurpleInventory references PurpleProduct.
- A seller can supply multiple beauty products.
- A product can be associated with inventory records.



3. Table Creation

3.1 Create Seller Table:

```
SQL> CREATE TABLE SellerDetails (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Email VARCHAR2(100) UNIQUE,  
5     Phone VARCHAR2(15),  
6     Address VARCHAR2(200),  
7     City VARCHAR2(50)  
8 );
```

Table created.

3.2 Create Inventory Table:

```
SQL> CREATE TABLE inventory_details (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER,  
4     Seller_ID NUMBER,  
5     Stock_Quantity NUMBER NOT NULL,  
6     Stock_Status VARCHAR2(20) NOT NULL,  
7     Last_Updated DATE NOT NULL,  
8     FOREIGN KEY (Seller_ID) REFERENCES seller_details(Seller_ID)  
9 );
```

Table created.

4. Seller Data Insertion:

```

SQL> INSERT INTO seller_details
2 VALUES (1, 'Purple Beauty Store', 'purplebeauty@gmail.com',
3 '9876543210', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO seller_details
2 VALUES (2, 'Purple Cosmetics', 'purplecosmetics@gmail.com',
3 '9876543211', 'Bangalore');

1 row created.

SQL>
SQL> INSERT INTO seller_details
2 VALUES (3, 'Purple Beauty Hub', 'purplehub@gmail.com',
3 '9876543212', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO seller_details
2 VALUES (4, 'Purple Care', 'purplecare@gmail.com',
3 '9876543213', 'Madurai');

1 row created.

SQL>
SQL> INSERT INTO seller_details
2 VALUES (5, 'Purple Glow', 'purpleglow@gmail.com',
3 '9876543214', 'Salem');

1 row created.

```

5. Inventory Data Insertion:

```

SQL> INSERT INTO inventory_details
2 VALUES (101, 101, 1, 50, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO inventory_details
2 VALUES (102, 102, 2, 30, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO inventory_details
2 VALUES (103, 103, 3, 0, 'Unavailable', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO inventory_details
2 VALUES (104, 104, 4, 25, 'Available', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO inventory_details
2 VALUES (105, 105, 5, 0, 'Unavailable', SYSDATE);

1 row created.

```

6. Seller Product Information:

6.1 Display Seller and Product Details:

```

SQL> SELECT
2      s.Seller_ID,
3      s.Seller_Name,
4      i.Product_ID,
5      i.Stock_Quantity,
6      i.Stock_Status
7 FROM seller_details s
8 JOIN inventory_details i
9 ON s.Seller_ID = i.Seller_ID;

```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
1
Purple Beauty Store
101          50 Available

2
Purple Cosmetics
102          30 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
3
Purple Beauty Hub
103          0 Unavailable

4
Purple Care

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
-----
104          25 Available

5
Purple Glow
105          0 Unavailable

```

7. Inventory Status :

7.1 Display Available Products:

```

SQL> SELECT *
2 FROM inventory_details
3 WHERE Stock_Status = 'Available';

```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
101	101	1	50	Available	18-SEP-26
102	102	2	30	Available	18-SEP-26
104	104	4	25	Available	18-SEP-26

```

SQL> SELECT
2 s.Seller_ID,
3 s.Seller_Name,
4 i.Product_ID,
5 i.Stock_Quantity,
6 i.Stock_Status
7 FROM seller_details s
8 JOIN inventory_details i
9 ON s.Seller_ID = i.Seller_ID
10 WHERE i.Stock_Status = 'Available';

```

SELLER_ID	SELLER_NAME	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
1	Purple Beauty Store	101	50	Available
2	Purple Cosmetics	102	30	Available

7.2 Display Unavailable Products:

```

SQL> SELECT *
2 FROM inventory_details
3 WHERE Stock_Status = 'Unavailable';

```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
103	103	3	0	Unavailable	18-SEP-26
105	105	5	0	Unavailable	18-SEP-26

```

SQL> SELECT
2 s.Seller_ID,
3 s.Seller_Name,
4 i.Product_ID,
5 i.Stock_Quantity,
6 i.Stock_Status
7 FROM seller_details s
8 JOIN inventory_details i
9 ON s.Seller_ID = i.Seller_ID
10 WHERE i.Stock_Status = 'Unavailable';

```

SELLER_ID	SELLER_NAME	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
3	Purple Beauty Hub	103	0	Unavailable
5	Purple Glow	105	0	Unavailable

8. Inventory Update:

Demonstrate updating stock information.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM PurpleProduct p
  7 JOIN inventory_details i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Stock_Status = 'Unavailable';
```

no rows selected

```
SQL> UPDATE inventory_details
  2 SET Stock_Quantity = 30,
  3     Stock_Status = 'Available',
  4     Last_Updated = '18-SEP-2026'
  5 WHERE Inventory_ID = 103;
```

1 row updated.

```
SQL> SELECT *
  2 FROM inventory_details
  3 WHERE Inventory_ID = 103;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
103	103	3	30	Available	18-SEP-26

9. Inventory Status Reports:

9.1 Complete Inventory Report:

```
SQL> SELECT
  2     i.Inventory_ID,
  3     p.Product_ID,
  4     p.Product_Name,
  5     s.Seller_ID,
  6     s.Seller_Name,
  7     i.Stock_Quantity,
  8     i.Stock_Status,
  9     i.Last_Updated
 10 FROM inventory_details i
 11 JOIN PurpleProduct p
 12 ON i.Product_ID = p.Product_ID
 13 JOIN seller_details s
 14 ON i.Seller_ID = s.Seller_ID;
```

no rows selected

```
SQL> SELECT * FROM inventory_details;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
101	101	1	50	Available	18-SEP-26
102	102	2	30	Available	18-SEP-26
103	103	3	30	Available	18-SEP-26
104	104	4	25	Available	18-SEP-26
105	105	5	0	Unavailable	18-SEP-26

9.2 Available and Unavailable Product Count:

```
SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Product_Count
  4 FROM inventory_details
  5 GROUP BY Stock_Status;
```

STOCK_STATUS	PRODUCT_COUNT
Available	4
Unavailable	1

9.3 Total Stock by Seller:

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Stock_Quantity) AS Total_Stock
  5 FROM seller_details s
  6 JOIN inventory_details i
  7 ON s.Seller_ID = i.Seller_ID
  8 GROUP BY s.Seller_ID, s.Seller_Name;
```

SELLER_ID

SELLER_NAME

TOTAL_STOCK

1
Purple Beauty Store
50

2
Purple Cosmetics
30

SELLER_ID

SELLER_NAME

TOTAL_STOCK

3
Purple Beauty Hub
30

4
Purple Care

9.4 Products with Zero Stock:

This is particularly useful for demonstrating unavailable products.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM inventory_details i
  8 JOIN PurpleProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN seller_details s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Quantity = 0;
```

no rows selected

```
SQL> SELECT *
  2 FROM inventory_details
  3 WHERE Stock_Quantity = 0;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
105	105	5	0	Unavailable	18-SEP-26